

Sunline Endeavour — Welding Consumables Web Application

Code audit, competitive research, architecture, and implementation plan

1. What you currently have

You uploaded **two competing codebases** for the same company.

1.1 Version A — `Index.html` (legacy, 1,477 lines)

Aspect	Finding
Structure	Single file, all CSS in an inline <code><style></code> block
Dead code	Lines 1–716 are entirely commented out — an earlier draft under a different company name (<i>ASA Industry</i> , Tiruchirapalli, different founders). The live document starts at line 718.
Branding	Title says "Sunline Endavour" (typo), body copy is all ASA Industry
Contact form	Posts to <code>https://formspree.io/f/YOUR_FORM_ID</code> — placeholder ID, the form silently fails
Content quality	Product grid has duplicated cards — "Stainless Steel Electrodes" twice, "Hardfacing Electrodes" twice, verbatim
Images	Hotlinked from Unsplash CDN mixed with local <code>./images/</code>
Sections	home, about, consumables, properties, contact

1.2 Version B — `index.html` + `style.css` + `script.js` (current, ~2,900 lines)

Significantly better engineering:

- Proper separation of concerns; BEM-ish class naming (`nav__link`, `product-card__title`)
- A real design-token layer in `:root` (26 custom properties: palette, gradients, type, spacing, transitions, shadows, radii)
- 18 media queries — responsive was actually considered
- `IntersectionObserver` scroll reveals, `requestAnimationFrame`-throttled parallax, debounce helper, lazy-load scaffolding
- Product modal driven by a `productData` object
- WhatsApp float with a bilingual EN/தமிழ் prefilled message
- OG tags, meta description, preconnect to Google Fonts

1.3 `send_email.php` — orphaned and unsafe

Nothing in Version B posts to it, and Version A posts to Formspree instead. So it's dead code — but if you ever wire it up, these are blocking defects:

#	Defect	Severity
1	Mail header injection. <code>\$name</code> is interpolated straight into <code>From: \$name</code> <code><\$email></code> . <cite index="61-1"> Without proper filtering, an attacker inserts a value after the colon separated with a CRLF character (<code>\r\n</code>), and some mail servers also accept bare <code>\n</code> to set <code>Bcc:</code> / <code>Cc:</code> headers and send spam through your form. </cite> <code>FILTER_SANITIZE_STRING</code> strips tags — it does not strip newlines.	Critical
2	<code>FILTER_SANITIZE_STRING</code> is <cite index="58-1"> deprecated as of PHP 8.1.0 and emits a deprecation warning; the official PHP documentation recommends <code>htmlspecialchars</code> instead </cite> . <cite index="59-1"> It was deprecated because its behaviour varied drastically depending on optional flags and it failed against modern XSS vectors. </cite>	High
3	<code>From:</code> header uses the <i>visitor's</i> address on <i>your</i> server → SPF/DKIM/DMARC alignment fails → mail is spam-folded or rejected outright. Correct pattern: <code>From:</code> = your own domain, <code>Reply-To:</code> = the visitor.	High
4	No honeypot, no timestamp check, no rate limit, no CAPTCHA → bot spam within days of going live	High
5	Unauthenticated <code>mail()</code> rather than authenticated SMTP	Medium
6	Recipient is a personal Gmail address hardcoded in source	Medium
7	Plain-text <code>echo</code> responses, no JSON contract for a JS-driven form	Low

1.4 Cross-cutting problems

1. Filename case collision. `Index.html` and `index.html` differ only in case. This works on ext4, **breaks on Windows and default macOS** (case-insensitive), and Git will produce phantom conflicts. Which one Apache serves depends on `DirectoryIndex` order. This must be resolved before anything else.

2. No single source of truth for product data. Specs live in the HTML markup *and* again in `productData` in `script.js`. The moment someone edits one, they drift. With ~50 SKUs this becomes unmaintainable.

3. No lead capture in the current version. Version B has contact *cards* and a WhatsApp button — but no form at all. Given that <cite index="36-1">the primary conversion event for industrial equipment is the RFQ, not a cart checkout, yet many industrial websites treat the quote form as an afterthought rather than a carefully optimized conversion point</cite>, this is the largest commercial gap in the build.

4. Placeholder content shipped as if real. `123 Industrial Estate`, social links all `href="#"`, `© 2024`, a WhatsApp number commented `// Test number`, a catalogue PDF path that may not exist, decorative SVG placeholders where factory photos should be.

5. Single-page architecture caps SEO hard. One URL means one page that can rank. Fifty products deserve fifty indexable, linkable, schema-marked pages.

6. Accessibility gaps. Modal has no focus trap, no `Escape` handler, no `aria-modal`; product cards use `<article onclick=...>` so they're unreachable by keyboard; images lack `width`/`height` (layout shift).

7. No engineering infrastructure. No `package.json`, no build, no linting, no tests, no CI, no environment config, no image pipeline.

2. Research: what the market is doing

2.1 How industrial buyers actually behave

`<cite index="7-1">`B2B industrial buyers complete a large majority of their supplier research online before making first contact with any vendor. They evaluate your website silently — and decide within the first 60 to 90 seconds whether you are worth pursuing.`</cite>`

`<cite index="36-1">`A plant engineer cares about specs and tolerances, an operations director cares about throughput and maintenance, and a procurement manager cares about price, lead time, and warranty terms — your website has to serve all three, often within the same product page. Industrial buyers resist early contact with sales and want to self-educate extensively; if your site forces "contact us for more information" before providing adequate technical detail, many leave and research a competitor whose site gives them what they need.`</cite>`

Implication: publish the full technical data openly. Don't gate the datasheets.

2.2 Information architecture

`<cite index="3-1">`Product navigation needs to be designed around technical categories rather than marketing language — engineers and procurement teams rarely browse casually, they arrive looking for specific product types, performance specs, or compliance details.`</cite>`

For welding consumables, that means the taxonomy is: **process → alloy family → AWS/IS classification → diameter**. Not "Our Consumables."

2.3 Trust signals

`<cite index="4-1">`Manufacturing is a trust business — buyers vet you before they ever pick up the phone. ISO certifications, industry experience, and key capabilities should be visible on the homepage, not buried three clicks deep on an About page nobody reads.`</cite>`

`<cite index="5-1">`Buyers need to know you are reliable: the site must prominently display ISO certifications, safety awards, association memberships, and client testimonials.`</cite>`

2.4 Discovery tools

`<cite index="6-1">`Common traps that limit industrial website performance: overly technical copy with no UX support; poor product discovery tools, where outdated PDFs and long dropdowns aren't enough and customers give up without filters or smart

search; and lack of mobile readiness, which frustrates field reps and decision makers on the go.</cite>

2.5 The RFQ is the money page

<cite index="35-1">An RFQ form differs from a generic contact form because it collects product-specific, transaction-ready detail instead of a vague inquiry. A visitor filling one out is telling you what they want to buy, in what quantity, and often by when.</cite>

<cite index="36-1">Offer a "quick quote" option alongside a detailed RFQ. A two-field quick quote form (part and quantity) captures early-stage interest that a 12-field form would kill — the website's job is to get the conversation started, not to qualify the lead to completion before the first human interaction. Guided product selectors that ask a short series of questions solve the hesitation buyers feel when they can't determine the right configuration from a static page.</cite>

<cite index="34-1">One CTA across every page is easier to manage, but it also hides buying intent instead of capturing it. An engineer who spends six minutes on a product page, checks specs, opens a PDF, then leaves without filling out your quote form was not a dead end — the ask was too big for that stage. Shorter conversion paths such as spec-sheet requests work better than bottom-of-funnel quote forms for that visitor.</cite>

Implication: three conversion tiers, not one.

- **Tier 1 (low friction):** download datasheet / catalogue — no form, or email-only
- **Tier 2 (medium):** quick price check — product + quantity + phone
- **Tier 3 (high):** full RFQ — multi-item, delivery location, timeline, application details

2.6 What competitors actually ship

Elga (ITW Welding) — the closest functional benchmark. Their site carries <cite index="23-1">a consumable calculator that takes joint type (fillet, V, X, K, U joint), deposition coefficient (1.0 STD, 0.70 SMAW, 0.87 FCAW, 0.98 GMAW/GTAW/SAW), weld metal density by alloy family (7.80 C and C-Mn steel, 8.30 nickel base, 7.90 stainless, 2.70 aluminium), groove angle and weld length</cite>, alongside <cite index="23-1">a download centre, certifications, PDS/MSDS/DoP and approvals, a technical enquiry route, and a customer login</cite>.

D&H Sécheron (India) — <cite index="26-1">organises around conventional consumables vs. reclamation consumables, plus reclamation services, case studies, blog, application articles, and education articles</cite>, and leans on <cite index="26-1">customer testimonials citing measured outcomes ("cost of reclamation has been less as well as the life has increased twice") and named mega-project case studies such as the Mumbai Trans Harbour Link</cite>. They emphasise <cite index="27-1">a widespread dealer network</cite>.

Lincoln Electric — catalogues consumables by <cite index="16-1">stick electrodes, MIG wires and TIG cut lengths, metal-cored, flux-cored (self- and gas-shielded), submerged arc, stainless and high alloy, hardfacing, aluminium — cross-indexed against industries served: automotive, general fabrication, heavy fabrication, maintenance and repair, offshore, pipeline, power generation, shipbuilding, structural</cite>. Two axes: *what it is* and *what you're building*.

A market gap worth owning: the most-circulated document in the Indian welding trade is [a brand cross-reference comparing over 90 electrodes across D&H Sécheron, ADOR Welding and ESAB India, giving the AWS coding and the equivalent brand product names, covering mild steel, low alloy, stainless, hardfacing, nickel alloys, cast iron and cellulosic](#) — and it lives on Scribd as a PDF. Publishing an interactive, indexable equivalence table on your own domain captures that entire search stream.

2.7 Technical SEO / answer-engine readiness

Search engines need to understand what you manufacture: proper schema markup for products, fast loading speeds (Core Web Vitals), and secure HTTPS. Technical SEO is the foundation of visibility — without it, even the most beautiful industrial site remains unseen.

The 2026 pillars are procurement-officer-focused UX, technical performance (Core Web Vitals compliance, sub-2-second LCP, mobile optimisation), and AEO-ready content architecture — FAQ sections, JSON-LD schema markup, and answer-first content structure — so the site is visible in both traditional Google search and AI answer engine results.

3. Feature set for Sunline Endeavour

Prioritised. Tier 1 is the launch scope; Tier 3 turns a website into an actual web application.

Tier 1 — Launch (must have)

Feature	Why
Per-product pages at real URLs (<code>/products/sle-7018</code>)	50 indexable pages instead of 1
Technical taxonomy — process → alloy family → AWS class → diameter	Matches how engineers search
Faceted product finder — filter by AWS class, IS code, alloy family, welding position, current type (AC/DC±), diameter, industry	Replaces the "long dropdown / PDF dump" failure mode
Full spec tables per product — classification & coding (AWS/ISO/IS), chemical composition, mechanical properties (UTS, YS, elongation, impact), welding current range per diameter, positions, redrying schedule, packing data	This is the product for a technical buyer
Datasheet + MSDS download per SKU , plus master catalogue	<code><cite index="23-1"></code> PDS, MSDS and DoP downloads are table stakes in this category <code></cite></code>
Three-tier conversion: datasheet download → quick price check → full RFQ	Section 2.5
RFQ basket — add multiple SKUs and sizes, submit once	Real orders are multi-line
Trust bar above the fold — ISO 9001, BIS/IS licence numbers, AWS conformance, years in operation, plant photography	Section 2.3
Real contact data + embedded map + WhatsApp deep link carrying product context	India-specific: WhatsApp converts
Bilingual EN / தமிழ்	Your existing script already gestures at this
JSON-LD: Organization, Product, BreadcrumbList, FAQPage	Section 2.7
Core Web Vitals budget enforced in CI	Section 2.7

Tier 2 — Differentiators (high ROI, build next)

Feature	Why
Electrode consumption & cost calculator	Direct analogue to Elga's tool. Joint geometry + length + deposition efficiency → kg of weld metal, rod count, cost, packing units. Highest-value lead magnet in this category — a buyer who calculates 40 kg of E7018 is one click from an RFQ.
Electrode selector wizard	Base metal + thickness + joint + position + service condition + power source → recommended SKU. <cite index="36-1">Guided selectors remove the hesitation of guessing the right configuration.</cite>
Brand cross-reference / equivalence finder	Section 2.6 gap. Enter E7018 or a competitor brand name → your equivalent. Enormous long-tail SEO.
AWS classification decoder	Interactive breakdown of E-70-1-8 : what each digit means (tensile strength, position, coating & current). Genuine educational utility, strong AEO asset.
Dealer / distributor locator	<cite index="27-1">Dealer network is a headline credential in this market.</cite>
Case studies with measured outcomes	<cite index="26-1">D&H's testimonials quantify results.</cite>
Technical articles / welding knowledge hub	AEO surface area

Tier 3 — The application layer

Feature	Why
Batch test-certificate lookup + QR on cartons	Every consumable buyer eventually asks for the TC for a batch. Print a QR on the carton → scans to a public page with the mill test certificate PDF, batch number, manufacture date, and heat data. Nobody in the Indian market does this well. It is the single most defensible feature on this list.
Distributor / customer portal	<cite index="6-1">Robust B2B sites include customer portals, tiered pricing, ERP or inventory integration, and role-based permissions with custom pricing rules.</cite> Login → contract price list, order history, dispatch/LR tracking, downloadable TCs, reorder.
Admin RFQ console	Inbox, status pipeline (new → quoted → won/lost), assignment, PDF quote generation, response SLA tracking
Stock / lead-time indicator per SKU	Removes the top pre-RFQ question
WhatsApp Business API	Automated RFQ acknowledgement + quote delivery
ERP / Tally integration	Indian SMB reality — the accounts team is already in Tally
Product-level analytics	Which datasheets were downloaded before which RFQs — feeds sales prioritisation

4. Architecture

4.1 Framework decision

The site is roughly **80% content** (catalogue, specs, datasheets, SEO pages) and **20% application** (RFQ, portal, admin). That ratio drives the choice.

<cite index="52-1">For content-first projects Astro ships far less JavaScript, scores higher on Lighthouse and loads faster; Next.js dominates authenticated, data-dense apps with Server Components, Server Actions and edge middleware. A useful pattern in 2026 is the hybrid stack: marketing site and docs in Astro for speed and SEO, app subdomain in Next.js for the authenticated surface.</cite> <cite index="52-1">Agencies increasingly default to Astro for campaign sites — instant LCP, trivial CDN hosting, and the freedom to drop in a React or Vue widget where a form or calculator is needed.</cite>

That last clause describes this project exactly: a static catalogue with a handful of interactive islands (calculator, selector, RFQ basket).

Recommendation: Astro 5 for the public site, React islands for interactivity, a small Hono API + Postgres for RFQ/portal/admin.

Alternative worth considering: if you'd rather run one framework end to end and you weight the Tier 3 portal heavily, Next.js App Router does all of it in a single codebase at

the cost of a heavier baseline bundle on catalogue pages. Both are defensible. Pick Astro if the catalogue matters most; pick Next.js if the portal matters most. Don't run both unless the portal grows large enough to justify a separate deploy.

```

sunline-endeavour/
├─ AGENTS.md                      # agent instructions (see prompt doc)
├─ package.json                  # pnpm workspace
├─ apps/
│   └─ web/                      # Astro 5 – public site
│       └─ src/
│           └─ pages/
│               └─ index.astro
│               └─ products/
│                   └─ index.astro      # faceted finder
│                   └─ [slug].astro    # SSG per SKU
│               └─ applications/[industry].astro
│               └─ tools/
│                   └─ consumption-calculator.astro
│                   └─ electrode-selector.astro
│                   └─ cross-reference.astro
│               └─ quality/            # ISO, certs, test
certificates
│   └─ batch/[batchNo].astro        # SSR – TC lookup
│   └─ distributors.astro
│   └─ knowledge/[...slug].astro
│   └─ contact.astro
│   └─ ta/                          # Tamil locale tree
│       └─ components/
│           └─ islands/              # React – hydrated
│               └─ ProductFinder.tsx
│               └─ ConsumptionCalculator.tsx
│               └─ ElectrodeSelector.tsx
│               └─ CrossReference.tsx
│               └─ RfqBasket.tsx
│               └─ QuickQuoteForm.tsx
│               └─ ...                # .astro – zero JS
│       └─ layouts/
│       └─ styles/tokens.css
│       └─ lib/schema.ts             # JSON-LD builders
│   └─ astro.config.mjs
├─ api/                           # Hono on Node
│   └─ src/routes/{rfq,quote,batch,portal,admin}.ts
├─ packages/
│   └─ content/                    # SINGLE SOURCE OF TRUTH
│       └─ products/*.yaml
│       └─ cross-reference.yaml
│       └─ distributors.yaml
│       └─ certifications.yaml
│       └─ schema.ts               # Zod – validated at build, build
fails on drift
│   └─ db/                         # Drizzle schema + migrations
│   └─ ui/                         # shared tokens + primitives

```

```
└─ assets/
  └─ datasheets/           # per-SKU PDF
  └─ msds/
  └─ certificates/
```

4.3 Content model (`packages/content/products/sle-7018.yaml`)

yaml

```
slug: sle-7018
name: SLE-7018
family: low-hydrogen
process: SMAW
tagline: Low hydrogen iron-powder electrode for critical structural work
classification:
  aws: E7018
  iso: "ISO 2560-A E 42 4 B 42 H5"
  is: "IS 814 EB5426H3JX"
coating: basic-low-hydrogen
current: [DC+, AC]
positions: [F, H, V-up, OH]
redry: { temp_c: 300, hold_hours: 2 }
chemistry:           # weld metal, %
  - { element: C, max: 0.08 }
  - { element: Mn, min: 0.90, max: 1.40 }
  - { element: Si, max: 0.60 }
mechanicals:
  uts_mpa: { min: 490 }
  yield_mpa: { min: 400 }
  elongation_pct: { min: 22 }
  impact_j: { value: 47, temp_c: -30 }
sizes:
  - { dia_mm: 2.50, len_mm: 350, amps: [60, 90], pcs_per_pkt: 100, kg_per_c
  - { dia_mm: 3.15, len_mm: 450, amps: [90, 130], pcs_per_pkt: 80, kg_per_c
  - { dia_mm: 4.00, len_mm: 450, amps: [130, 180], pcs_per_pkt: 50, kg_per_c
applications: [structural, pressure-vessel, shipbuilding, bridges]
industries: [construction, heavy-fabrication, power]
datasheet: /assets/datasheets/sle-7018.pdf
msds: /assets/msds/sle-7018.pdf
equivalents:
  - { brand: "D&H Sécheron", product: "Supratherm" }
  - { brand: "Ador", product: "Tenalloy 18" }
  - { brand: "ESAB India", product: "Vordian 7018" }
```

Zod validates every file at build time. A malformed spec fails `pnpm build` rather than shipping a wrong amperage range to a customer. Product data stays in Git — reviewable, diffable, no CMS to pay for or secure. Move it into Postgres + an admin UI only when a non-developer needs to edit it daily.

4.4 Database (transactional data only, v1)

```
rfqs(id, ref_no, company, contact_name, email, phone, country, state,
     message, source_page, utm_*, status, assigned_to, created_at)
rfq_items(id, rfq_id, product_slug, dia_mm, qty, unit)      -- unit: kg |
cartons
quick_quotes(id, product_slug, qty, phone, created_at)
downloads(id, product_slug, doc_type, email?, ip_hash, created_at)
batches(batch_no PK, product_slug, mfg_date, expiry, tc_pdf_url, heat_no)
distributors(id, name, city, state, lat, lng, phone, whatsapp, tier)
-- Tier 3
organizations(id, name, gstin, credit_terms)
users(id, org_id, email, password_hash, role)
price_lists(org_id, product_slug, dia_mm, rate, valid_from, valid_to)
orders(...), dispatches(...)
```

4.5 Forms and email — replacing `send_email.php`

Delete the PHP entirely. The API route does:

1. **Zod validation** server-side (never trust the client)
2. **Honeypot field** + minimum time-on-form check (bots submit in <2s)
3. **Cloudflare Turnstile** (free, no puzzle for humans)
4. **Rate limit** per IP — 5 submissions / 10 min
5. **Strip `\r` and `\n` from every value that touches a mail header** — the fix for defect #1
6. Send via **Resend / Postmark / Brevo / SES** with `From: rfq@sunlineendeavour.com`, `Reply-To: <visitor>` — the alignment fix for defect #3
7. Write the RFQ to Postgres *before* attempting to send, so a mail failure never loses a lead
8. Auto-acknowledge to the customer with the RFQ reference number
9. Return JSON `{ ok, refNo }`

Configure **SPF, DKIM and DMARC** on the domain — without them the acknowledgement mails land in spam regardless of provider.

If you're locked to shared PHP hosting, keep PHP but rewrite it: replace

`FILTER_SANITIZE_STRING` with `htmlspecialchars(..., ENT_QUOTES, 'UTF-8')`, run every header value through `preg_replace('/[\r\n]+/', ' ', $v)`, set `From:` to your own domain, use PHPMailer with authenticated SMTP, and add the honeypot + rate limit. Then the current file becomes safe.

4.6 Performance budget (enforced in CI, build fails on breach)

Metric	Target
LCP (mobile, 4G throttled)	< 2.0 s
CLS	< 0.1
INP	< 200 ms
JS shipped, catalogue page	< 40 KB gzipped
JS shipped, homepage	< 25 KB gzipped
Lighthouse Performance / SEO / A11y	≥ 95 each

Techniques: self-host fonts with `font-display: swap` and subset to Latin + Tamil; AVIF/WebP via Astro `<Image>` with explicit dimensions; islands hydrate `client:visible` not `client:load`; no analytics script above the fold.

4.7 SEO / AEO implementation

- One canonical URL per SKU; `hreflang` pairs for `en` / `ta`
- `Product` JSON-LD with `sku`, `mpn`, `brand`, `manufacturer`, and each spec as an `additionalProperty` — this is how an AI answer engine reads your amperage table
- `FAQPage` on every product page ("What current does SLE-7018 run on?", "What is the redrying temperature?") — answer-first, one paragraph each
- `BreadcrumbList`, `Organization` with `hasCredential` for ISO
- Generated `sitemap.xml`, explicit `robots.txt`
- Auto-generated OG images per product at build

4.8 Design direction

Your current palette is a generic corporate blue. The subject has a far more specific visual world to draw from — arc light, flux coating, slag, mill-stencil lettering on cartons, the AWS code itself.

- **Palette:** graphite ground (`#1B1D21`), arc-white (`#F2F4F5`), molten amber accent (`#E8892B`) taken from the weld pool, flux grey (`#8A8F94`), and a single cool slag-blue (`#3D6E8C`) for links only. Amber is the accent because it's the colour of the work; blue becomes secondary.
- **Type:** a condensed grotesk for display (industrial signage vernacular — e.g. Archivo Condensed), a neutral text face for body, and a **tabular-figure monospace for all spec tables**. Non-negotiable: amperage ranges and composition percentages must align vertically or the tables are unreadable.
- **Signature element:** the interactive **AWS classification decoder** on each product page — `E · 70 · 1 · 8` broken into four tappable segments that expand to explain tensile strength, welding position, and coating/current. It encodes something true about the content rather than decorating it, it's unique to this industry, and it doubles as your best AEO asset.

- **Restraint:** spend the boldness there. Everything else stays quiet — the audience is a procurement manager on a phone in a plant, not a design juror. Keyboard focus visible, `(prefers-reduced-motion)` respected, tap targets ≥ 44 px.

4.9 Hosting

Layer	Choice
Static site	Cloudflare Pages or Netlify (free tier covers this comfortably)
API	Fly.io / Render / Railway — or the same VPS behind Caddy
Postgres	Neon or Supabase
PDFs / images	Cloudflare R2 or S3 + CDN
Email	Resend or Brevo (Brevo has better India deliverability)
Analytics	Plausible or Umami — cookieless, ~1 KB

5. Implementation plan

Each phase has an **exit criterion**. Don't start the next phase until it's met.

Phase 0 — Triage (½ day)

1. Choose Version B as canonical. Delete `(Index.html)` and `(send_email.php)` in the same commit — record why in the message.
2. `(git config core.ignorecase false)`; verify no case-colliding paths remain.
3. Initialise npm workspace, TypeScript strict, ESLint, Prettier, Husky pre-commit.
4. Inventory every real asset: which datasheets exist, which photos exist, which specs are verified vs. invented.

Exit: one `(index.html)` in the tree, `(npm lint)` passes, a written list of missing content.

Phase 1 — Foundation (3-4 days)

Astro scaffold; port the 26 design tokens from `(style.css)` into `(tokens.css)` and revise per §4.8; base layout, header, footer, nav; self-hosted fonts; homepage rebuilt from Version B's markup; deploy preview live.

Exit: homepage on a preview URL, Lighthouse ≥ 95 across the board, zero content regressions vs. Version B.

Phase 2 — Catalogue (4-5 days)

Zod content schema; migrate all SKUs from `(productData)` + HTML into YAML (this is where the duplicate-card problem gets resolved for good); `(/products/[slug])` SSG; faceted `(ProductFinder)` island; spec tables; datasheet links; AWS decoder component; `(Product)` + `(Breadcrumb)` JSON-LD; sitemap.

Exit: every SKU has its own URL, all facets filter correctly, Rich Results Test passes on three sampled product pages, build fails if a YAML file is malformed.

Phase 3 — Conversion (3–4 days)

Hono API + Postgres + Drizzle; RFQ endpoint with the full §4.5 hardening; `RfqBasket` island persisting to `sessionStorage`; `QuickQuoteForm` (2 fields) on every product page; datasheet-download tracking; transactional email both directions; WhatsApp deep link prefilled with the viewed SKU; contact page with real data.

Exit: an RFQ submitted from a phone lands in Postgres *and* both inboxes within 30 s; three header-injection payloads and a honeypot submission are all rejected; SPF/DKIM/DMARC pass on `mail-tester.com` with $\geq 9/10$.

Phase 4 — Tools (4 days)

Consumption calculator (fillet / V / X / K / U joints, deposition efficiency presets per process, weld-metal density per alloy family, output in kg + rod count + cartons + cost, with a "Request quote for this quantity" button wired straight into the RFQ basket); electrode selector wizard; cross-reference table with its own indexable page per AWS class.

Exit: calculator output verified against three hand-worked examples; every calculator result offers a one-tap RFQ; cross-reference page indexed for at least one competitor brand term.

Phase 5 — Trust, SEO, i18n (3 days)

Quality page with ISO/BIS scans; certifications; case studies; knowledge-hub articles; FAQ blocks + `FAQPage` schema; Tamil locale with `hreflang`; OG image generation; real factory photography replacing placeholder SVGs.

Exit: all placeholder copy gone (grep for `123 Industrial`, `href="#"`, `2024`, `YOUR_FORM_ID`); Tamil homepage and one product page complete; Search Console clean.

Phase 6 — Application layer (2 weeks, optional)

Batch TC lookup + QR generation; distributor locator; admin RFQ console with auth; stock indicator; portal login with org price lists.

Exit: scanning a printed carton QR opens the correct test certificate on a phone in under 3 s.

Phase 7 — Launch (2 days)

CI: lint + typecheck + build + Lighthouse budget + Zod content validation on every PR. Analytics, uptime monitoring, error tracking, backups, 301s from old URLs, `robots.txt`, Search Console + Bing submission.

Exit: green CI on `main`, monitoring alerting to a real phone, backup restore tested once.

Total to a strong launch (Phases 0–5, 7): ~3 weeks of focused solo work.

6. Risk register

Risk	Mitigation
Fabricated technical specs. Some values in the current code look like placeholders. Publishing a wrong amperage range or composition is a liability, not a bug.	Every spec must be signed off against the actual QC datasheet before it ships. Mark unverified fields <code>null</code> and hide them rather than guessing.
Missing assets (catalogue PDF, factory photos, ISO scans)	Phase 0 inventory; block Phase 5 until supplied
Form spam	Turnstile + honeypot + rate limit from day one, not retrofitted
Scope creep into the portal before the catalogue earns traffic	Phases 0–5 ship first. The portal is worthless without customers using the site.
Email deliverability from a new domain	Warm up gradually; configure SPF/DKIM/DMARC in Phase 3, not at launch